

THE HUMP BETWEEN TWO CLOSE ZEROS: RANDOM UNITARY MATRICES AND THE RIEMANN ZETA FUNCTION

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ABSTRACT. Let M be the maximum of $|\Lambda_N|$, the characteristic polynomial of a Haar unitary $N \times N$ matrix, on the arc between two consecutive eigenangles at distance s . For fixed N we prove that, conditionally on $s < \varepsilon$, the pair $(s/\varepsilon, M/s^2)$ converges in law as $\varepsilon \rightarrow 0$ to independent variables $(U^{1/3}, X_{N-2}/4)$, where U is uniform and X_{N-2} is $|\Lambda_{N-2}(1)|$ under the $|\Lambda_{N-2}(1)|^4$ -tilted Haar measure. Such gaps occur at rate $N^2(N^2 - 1)\varepsilon^3/(72\pi)$, and by the product formula of Bourgade, Hughes, Nikeghbali and Yor all moments and log-cumulants of the hump are explicit; in unfolded units the log-hump has mean $\rightarrow 2 \log \pi + 2\gamma - 10/3$ and a negative third cumulant $\rightarrow -0.1271$. For the Riemann zeros we cast the analogue as a tilt ladder, conditioning on $b = 0, 1, 2$ zeros, and test it on zeros at heights $L = \log(t/2\pi) \approx 9$ – 22 . On the first two rungs, $\log |\zeta(\frac{1}{2} + it)|$ and $\log |\zeta'(\rho)|$, we derive the variance and the third cumulant at finite height from the ratios conjecture, using three- and four-point averages of ζ'/ζ and the density of zeros; the identity term reproduces the constant of Fazzari and Gerspach. Without free parameters the four predictions match the data at all seven heights (χ^2 between 2.5 and 11.6 for 7 points), while the random-matrix model with the Keating–Snaith arithmetic factor, which is right in the limit, fails at these heights (χ^2 up to 16 826). Pre-registered tests also show that midpoints of close pairs lock to the phases of the prime waves about four times as strongly as single zeros.

1. INTRODUCTION

Let $\gamma_n < \gamma_{n+1}$ be consecutive ordinates of zeros of the Riemann zeta function on the critical line, $\delta_n = \gamma_{n+1} - \gamma_n$, and let $M_n = \max_{\gamma_n \leq t \leq \gamma_{n+1}} |Z(t)|$ be the height of Hardy's function between them. The mean of M_n^2 over all gaps is known [13, 21]; the *joint* law of (δ_n, M_n) was studied numerically in [1], whose first open problem asked for the mean of M_n^2 conditioned on the gap. A reader asked the sharper question: in the regime of asymptotically small gaps, can the joint distribution of (δ_n, M_n^2) be determined?

For random unitary matrices the answer is yes, and exact. When two zeros nearly coincide, the hump between them is crushed by the square of the gap, and what is left is the height produced at the collision point by all the *other* zeros. For the characteristic polynomial of a Haar unitary matrix, conditioning two eigenvalues to merge at a point tilts the remaining $N - 2$ eigenvalues by $|\Lambda_{N-2}|^4$ at that point; under this tilt the value $|\Lambda_{N-2}|$ is a product of independent factors, so every moment and every cumulant of the logarithm is explicit.

Results for the unitary group. Let s be a gap between consecutive eigenangles of $U \in U(N)$ and M the maximum of $|\Lambda_N|$ on it (Section 2).

- *Theorem 1* (fixed N , $\varepsilon \rightarrow 0$). The expected number of gaps below ε is $N^2(N^2 - 1)\varepsilon^3/(72\pi)(1 + o(1))$; conditionally on $s < \varepsilon$ (for a gap chosen uniformly), $(s/\varepsilon, M/s^2) \Rightarrow (U^{1/3}, X_{N-2}/4)$ with U uniform on $[0, 1]$ and *independent* of X_{N-2} , the

value $|\Lambda_{N-2}(1)|$ under the $|\Lambda_{N-2}(1)|^4$ -tilted Haar measure. All moments $\mathbb{E}(M/s^2)^{2k}$ with $k > -5/2$ and all log-moments converge; the threshold is sharp.

- *Theorem 2.* X_n has the law of $\prod_{j=1}^n |1 - \xi_j|$ with independent, explicitly distributed ξ_j ; $\mathbb{E}X_n^{2k}$ is a ratio of Keating–Snaith moments and $\kappa_r(\log X_n) = \sum_{j=1}^n [\psi^{(r-1)}(j+4) - 2^{1-r}\psi^{(r-1)}(j+2)]$.
- *Corollary 3* (large N , unfolded units). The moments grow like N^{k^2} with the constant $\pi^{4k} 12 G(k+3)^2 / G(2k+5)$; the log-hump has mean $\rightarrow 2 \log \pi + 2\gamma - 10/3$, variance $\frac{1}{2} \log N + c_2 + o(1)$ with $c_2 = 65/18 + \gamma/2 - \pi^2/2$, third cumulant negative for every N and tending to -0.127077 , and is asymptotically Gaussian.
- Small gaps are invisible in the second moment: gaps below $\tilde{\varepsilon}$ in unfolded units carry a share of order $\tilde{\varepsilon}^7$ of $\mathbb{E} \sum_n M_n^2 \sim \frac{e^2-5}{2} N^2$, the constant being Winn’s [29] (Section S3 of the Supplementary material).

What is known. To our knowledge, the joint (gap, hump) law has not been written down before, but its ingredients are known. That conditioning the eigenvalues on points at a given location tilts the remaining points by powers of the characteristic polynomial is the Palm-measure structure of these ensembles (for the unitary group it is immediate from Weyl’s formula; see [17] for the log-gas setting). The Haar measure tilted by $|\det(1 - U)|^{2\delta}$ is the circular Jacobi ensemble [17, 8], which for $\delta \in \mathbb{N}$ is the CUE conditioned to have δ further eigenvalues at 1 [8, §1.1], and the product formula for $\det(1 - U)$ under Haar and tilted measures is due to Bourgade, Hughes, Nikeghbali and Yor [7] and Bourgade, Nikeghbali and Rouault [8]. Theorem 2 is a direct consequence of their results and is presented as such. The small gaps of the CUE were determined by Ben Arous and Bourgade [4] (Poisson limit at scale $N^{-4/3}$; Theorem 1(a) agrees with their intensity), and small gaps of other ensembles in [14, 15, 6]; these works concern the gaps, not the polynomial between the eigenvalues. Joint moments of the characteristic polynomial and its derivatives in the circular Jacobi ensemble have recently been related to Painlevé equations [5]; conditioning on coalescing eigenvalues is not considered there. The proof of Theorem 1 is elementary (Weyl’s formula and dominated convergence); the negative moments require a uniform domination, which follows from a one-line geometric inequality (Lemma B).

The zeta side. Conditioning on b zeros at a point tilts the rest of the spectrum by $|\Lambda|^{2b}$: $b = 0$ is the value distribution of $\log |\zeta|$ (Keating–Snaith [22]), $b = 1$ is $\log |\zeta'(\rho)|$ (Hughes–Keating–O’Connell [20]; see [9]), and $b = 2$ is the log-hump of close pairs studied here. Adding the Keating–Snaith arithmetic factor a_k to each rung gives a parameter-free prediction for the cumulants (Section 7), which we tested in pre-registered steps on Riemann zeros (Section 8). The arithmetic shift of the variance is nearly rung-independent and close to that of a_k , but the arithmetic shift of the third cumulant decreases from rung to rung at $L \approx 9$ –12 — a decrease that was predicted blind for $b = 2$ from the two lower rungs — and the ladder flattens as the height grows. On the first two rungs this finite-height behaviour is now accounted for without free parameters (Section 9): the ratios conjecture of Conrey, Farmer and Zirnbauer, evaluated with three- and four-point averages of ζ'/ζ that we derive (Appendix A) and weighted by the density of zeros, gives the variance and the third cumulant of $\log |\zeta(\frac{1}{2} + it)|$ and of $\log |\zeta'(\rho)|$ at each height, and all four match the data (Figure 1). The close-pair rung $b = 2$ remains open. A further pre-registered test finds that midpoints of close pairs lock to the phases of the prime waves about four times as strongly as single zeros, as a local-GUE heuristic predicts; this lock has the predicted size and does not by itself explain the decay along the

ladder. We report the failed parts of the predictions next to the successful ones. Nothing here bears on a proof of the Riemann Hypothesis.

Complements to the proofs, the second-moment computation, the full Monte Carlo record and the complete protocol and results of the pre-registered tests are in the Supplementary material, whose sections are numbered S1–S8.

2. SETUP AND NOTATION

Eigenangles. Let U be Haar distributed on $U(N)$, $N \geq 2$, with eigenvalues $e^{i\theta_1}, \dots, e^{i\theta_N}$, the labelled eigenangles $\theta = (\theta_1, \dots, \theta_N) \in \mathbb{T}^N$, $\mathbb{T} = [0, 2\pi)$, being exchangeable with density (Weyl's integration formula)

$$(1) \quad p_N(\theta) = \frac{1}{(2\pi)^N N!} \prod_{1 \leq j < k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^2.$$

The density is invariant under the rotation $\theta_j \mapsto \theta_j + \alpha$; almost surely the θ_j are distinct. For $n \geq 0$ we write $\mathbb{E}_n$ for the expectation under p_n ($\mathbb{E}_0$ is trivial) and

$$|\Lambda_n(e^{i\vartheta})| = \prod_{j=1}^n |e^{i\vartheta} - e^{i\theta_j}| = \prod_{j=1}^n 2 \left| \sin \frac{\vartheta - \theta_j}{2} \right|, \quad \Lambda_0 \equiv 1,$$

for the modulus of the characteristic polynomial $\det(e^{i\vartheta} - U)$; only the modulus is used.

Keating–Snaith moments. For $n \geq 1$ and real k ,

$$(2) \quad \mathcal{M}_n(k) := \mathbb{E}_n |\Lambda_n(1)|^{2k} = \prod_{j=1}^n \frac{\Gamma(j)\Gamma(j+2k)}{\Gamma(j+k)^2} \quad (k > -\tfrac{1}{2}),$$

and $\mathcal{M}_n(k) = \infty$ for $k \leq -\frac{1}{2}$ [22] (see also the proof of Theorem 1(c)); $\mathcal{M}_0 \equiv 1$. By rotation invariance $\mathbb{E}_n |\Lambda_n(e^{i\alpha})|^{2k} = \mathcal{M}_n(k)$ for every α . Telescoping the products gives

$$(3) \quad \begin{aligned} \mathcal{M}_n(2) &= \frac{(n+1)(n+2)^2(n+3)}{12}, \\ \mathcal{M}_n(3) &= \frac{(n+1)(n+2)^2(n+3)^3(n+4)^2(n+5)}{8640}. \end{aligned}$$

Gaps and humps. Order the eigenangles, $\theta_{(1)} < \dots < \theta_{(N)}$, put $\theta_{(N+1)} = \theta_{(1)} + 2\pi$, and let

$$s_n = \theta_{(n+1)} - \theta_{(n)}, \quad M_n = \max_{\theta_{(n)} \leq \vartheta \leq \theta_{(n+1)}} |\Lambda_N(e^{i\vartheta})|, \quad n = 1, \dots, N.$$

The unfolded gap is $\tilde{s}_n = N s_n / 2\pi$ (mean one). For a function G_n of the n -th gap we write

$$(4) \quad \langle G \rangle_\varepsilon := \frac{\mathbb{E} \sum_{n=1}^N \mathbf{1}\{s_n < \varepsilon\} G_n}{\mathbb{E} \#\{n : s_n < \varepsilon\}} = \mathbb{E}[G_J \mid s_J < \varepsilon],$$

where J is uniform on $\{1, \dots, N\}$ and independent of U : this is the law of a typical gap conditioned to be small.

The tilted law. For $n \geq 1$ let $\tilde{\mathbb{P}}_n$ be the Haar measure of $U(n)$ tilted by $|\Lambda_n(1)|^4/\mathcal{M}_n(2)$, $\tilde{\mathbb{E}}$ its expectation, and X_n the random variable $|\Lambda_n(1)|$ under $\tilde{\mathbb{P}}_n$. This is the circular Jacobi ensemble of [8, Def. 1.1] with $\beta = 2$ and $\delta = 2$: their weight $(1 - e^{-i\theta_j})^\delta(1 - e^{i\theta_j})^{\bar{\delta}}$ equals $|1 - e^{i\theta_j}|^{2\delta}$ for real δ . Thus

$$\tilde{\mathbb{E}} h(X_n) = \frac{\mathbb{E}_n[|\Lambda_n(1)|^4 h(|\Lambda_n(1)|)]}{\mathcal{M}_n(2)}.$$

We set $X_0 \equiv 1$. In unfolded units the limit object is $Y_N := (2\pi/N)^2 X_{N-2}/4$.

Special functions. $H_m^{(r)} = \sum_{i=1}^m i^{-r}$, $H_m = H_m^{(1)}$; $\psi = \Gamma'/\Gamma$ and $\psi^{(r)}$ its derivatives; γ is Euler's constant; G is the Barnes function, $G(z+1) = \Gamma(z)G(z)$, $G(1) = 1$; $\text{sinc } x = \sin x/x$. Cumulants are κ_r .

3. THE SMALL-GAP LIMIT

Theorem 1. Let $N \geq 2$ and $B_N := 2^{N-2}/4$.

- (i) For every configuration of distinct eigenangles and every n , $0 < M_n/s_n^2 \leq B_N$.
- (ii) Let $F : (0, 1] \times (0, B_N] \rightarrow \mathbb{R}$ be continuous, and suppose that $|F(u, y)| \leq C(1 + y^{2k_0})$ for some $C < \infty$ and some $k_0 \in (-\frac{5}{2}, 0]$. Then

$$(5) \quad \lim_{\varepsilon \downarrow 0} \varepsilon^{-3} \mathbb{E} \sum_{n=1}^N \mathbf{1}\{s_n < \varepsilon\} F\left(\frac{s_n}{\varepsilon}, \frac{M_n}{s_n^2}\right) = \frac{1}{2\pi} \int_0^1 u^2 \mathbb{E}_{N-2} \left[|\Lambda_{N-2}(1)|^4 F\left(u, \frac{|\Lambda_{N-2}(1)|}{4}\right) \right] du.$$

In particular:

- (a) $\mathbb{E} \#\{n : s_n < \varepsilon\} = \frac{N^2(N^2 - 1)}{72\pi} \varepsilon^3 (1 + o(1))$.
- (b) Under $\langle \cdot \rangle_\varepsilon$, $(s_J/\varepsilon, M_J/s_J^2) \Rightarrow (U^{1/3}, X_{N-2}/4)$ as $\varepsilon \rightarrow 0$, where U is uniform on $[0, 1]$ and independent of X_{N-2} . In the small-gap limit, gap and normalised hump are independent.
- (c) For every $k > -\frac{5}{2}$, $\langle (M/s^2)^{2k} \rangle_\varepsilon \rightarrow 4^{-2k} \tilde{\mathbb{E}} X_{N-2}^{2k} = 4^{-2k} \mathcal{M}_{N-2}(k+2)/\mathcal{M}_{N-2}(2)$, and for every $m > 0$, $\langle |\log(M/s^2)|^m \rangle_\varepsilon \rightarrow \tilde{\mathbb{E}} |\log(X_{N-2}/4)|^m$; the same holds for integer powers of $\log(M/s^2)$, so all cumulants of $\log(M/s^2)$ converge. If $N \geq 3$ and $k \leq -\frac{5}{2}$, then $\langle (M/s^2)^{2k} \rangle_\varepsilon \rightarrow \infty$.

The factor $|\Lambda_{N-2}(1)|^4$ in (5) is the product of the two factors $|\Lambda_{N-2}(e^{i\theta})|^2$ that the Vandermonde determinant attaches to each of the two merging eigenvalues; u^2 is their mutual repulsion. Ben Arous and Bourgade [4, Thm. 1.1] showed that the points $(N^{4/3}(\theta_{(n+1)} - \theta_{(n)}), \theta_{(n)})$ (gaps in angle units) converge to a Poisson process with intensity $\frac{1}{24\pi} u^2 du \frac{d\theta}{2\pi}$, so that the mean number of gaps below $xN^{-4/3}$ tends to $x^3/(72\pi)$; setting $\varepsilon = xN^{-4/3}$ in (a) gives the same leading term. The two limits are taken in different orders ($\varepsilon \rightarrow 0$ at fixed N here, $N \rightarrow \infty$ at fixed x there); the constants agree.

3.1. Two deterministic lemmas. Fix $s \in (0, 2\pi)$ and a configuration $\vartheta = (\vartheta_1, \dots, \vartheta_{N-2}) \in \mathbb{T}^{N-2}$ of the remaining angles, with $|\Lambda_{N-2}|$ built from ϑ . Put

$$\begin{aligned} M(s; \vartheta) &:= \max_{0 \leq t \leq s} |e^{it} - 1| |e^{it} - e^{is}| |\Lambda_{N-2}(e^{it})|, \\ W_s(\vartheta) &:= \prod_{j=1}^{N-2} |e^{i\vartheta_j} - 1|^2 |e^{i\vartheta_j} - e^{is}|^2, \\ A_s &:= \{\vartheta : \vartheta_j \notin (0, s) \text{ for all } j\}. \end{aligned}$$

$M(s; \vartheta)$ is the hump over the arc $[0, s]$ of the polynomial with zeros at 1, e^{is} and $e^{i\vartheta_j}$; W_s is the Vandermonde coupling between the pair and the rest; A_s says that the arc is a gap. For $N = 2$ all products over j are empty.

Lemma A (Hump bounds). *For every $s \in (0, 2\pi)$ and every ϑ ,*

$$(6) \quad \frac{4 \sin^2(s/4)}{s^2} |\Lambda_{N-2}(e^{is/2})| \leq \frac{M(s; \vartheta)}{s^2} \leq \frac{1}{4} \max_{0 \leq t \leq s} |\Lambda_{N-2}(e^{it})| \leq B_N.$$

Consequently $M(s; \vartheta)/s^2 \rightarrow |\Lambda_{N-2}(1)|/4$ as $s \downarrow 0$, for every ϑ ; and $M(s; \vartheta)/s^2 \geq \frac{1}{5} |\Lambda_{N-2}(e^{is/2})|$ for $0 < s \leq 1$.

Proof. For $t \in [0, s]$ the numbers $t/2$ and $(s-t)/2$ lie in $[0, \pi)$, so $|e^{it} - 1| = 2 \sin(t/2)$ and $|e^{it} - e^{is}| = 2 \sin((s-t)/2)$, and

$$q_s(t) := 4 \sin \frac{t}{2} \sin \frac{s-t}{2} = 2 \left[\cos \left(t - \frac{s}{2} \right) - \cos \frac{s}{2} \right].$$

Thus q_s is maximal at $t = s/2$, where it equals $2(1 - \cos \frac{s}{2}) = 4 \sin^2 \frac{s}{4}$; evaluating the product at $t = s/2$ gives the lower bound. Since $0 \leq \sin x \leq x$ for $x \in [0, \pi]$, $q_s(t) \leq t(s-t) \leq s^2/4$, which gives the middle bound, and $|e^{it} - e^{i\vartheta_j}| \leq 2$ gives the last one. As $s \downarrow 0$, $4 \sin^2(s/4)/s^2 = \frac{1}{4} \text{sinc}^2(s/4) \rightarrow \frac{1}{4}$ and both $|\Lambda_{N-2}(e^{is/2})|$ and $\max_{[0, s]} |\Lambda_{N-2}|$ tend to $|\Lambda_{N-2}(1)|$ by continuity. Finally, sinc is decreasing on $(0, \pi)$, so for $s \leq 1$, $4 \sin^2(s/4)/s^2 \geq 4 \sin^2(1/4) = 0.2448 \dots > \frac{1}{5}$. $\square$

Part (i) of Theorem 1 follows: after a rotation, the n -th gap is the arc $[0, s_n]$ with ϑ the other eigenangles, so $M_n = M(s_n; \vartheta)$; the upper bound is (6), and $M_n > 0$ because $M_n \geq |\Lambda_N|$ at the midpoint of the gap, which is not an eigenangle.

Lemma B (Domination). *For every $s \in (0, 2\pi)$ and every $\vartheta \in A_s$,*

$$(7) \quad W_s(\vartheta) \leq |\Lambda_{N-2}(e^{is/2})|^4.$$

Consequently, for $k \in (-\frac{5}{2}, 0]$, $0 < s \leq 1$ and $\vartheta \in A_s$,

$$(8) \quad W_s(\vartheta) \left(\frac{M(s; \vartheta)}{s^2} \right)^{2k} \leq 5^{-2k} |\Lambda_{N-2}(e^{is/2})|^{4+2k},$$

and $\mathbb{E}_{N-2} |\Lambda_{N-2}(e^{is/2})|^{4+2k} = \mathcal{M}_{N-2}(k+2) < \infty$.

The inequality (7) is index by index: for a point outside an arc, the product of its distances to the two endpoints of the arc is at most the square of its distance to the midpoint.

Proof. Let $\vartheta_j \notin (0, s)$ and let $x_j \in (-\pi, \pi]$ be the representative of $\vartheta_j - s/2$ modulo 2π . The points of $(0, s)$ are exactly those at circular distance less than $s/2$ from $s/2$, so $s/2 \leq |x_j| \leq \pi$. Since $|\sin(y/2)|$ is 2π -periodic,

$$|e^{i\vartheta_j} - 1| |e^{i\vartheta_j} - e^{is}| = 4 \left| \sin \frac{x_j + s/2}{2} \sin \frac{x_j - s/2}{2} \right| = 2 \left| \cos \frac{s}{2} - \cos x_j \right|,$$

by $2 \sin a \sin b = \cos(a - b) - \cos(a + b)$. Now $\cos \frac{s}{2} - \cos x_j = 2 \sin^2 \frac{x_j}{2} - 2 \sin^2 \frac{s}{4}$, and since $s/4 \leq |x_j|/2 \leq \pi/2$ and $\sin^2$ is increasing on $[0, \pi/2]$, $0 \leq \cos \frac{s}{2} - \cos x_j \leq 2 \sin^2 \frac{x_j}{2}$. Hence

$$|e^{i\vartheta_j} - 1| |e^{i\vartheta_j} - e^{is}| \leq 4 \sin^2 \frac{x_j}{2} = |e^{i\vartheta_j} - e^{is/2}|^2,$$

and squaring and multiplying over j gives (7). For (8), combine (7) with the lower bound $M/s^2 \geq \frac{1}{5} |\Lambda_{N-2}(e^{is/2})|$ of Lemma A (raised to the power $2k \leq 0$). The expectation equals $\mathcal{M}_{N-2}(k+2)$ by rotation invariance and is finite because $k+2 > -\frac{1}{2}$. $\square$

Configurations in which several ϑ_j approach the midpoint at once are included in the exact moment $\mathcal{M}_{N-2}(k+2)$; a self-contained check of its finiteness and the resulting uniform integrability of $(M_J/s^2)^{2k}$ are in Section S1.

3.2. Reduction to a two-point integral.

Lemma C. *Let $N \geq 2$, $0 < \varepsilon < 2\pi$, and let $g : (0, 2\pi) \times (0, \infty) \rightarrow [0, \infty]$ be measurable. Then*

(9)

$$\varepsilon^{-3} \mathbb{E} \sum_{n=1}^N \mathbf{1}\{s_n < \varepsilon\} g(s_n, M_n) = \frac{1}{2\pi} \int_0^1 u^2 \operatorname{sinc}^2 \left(\frac{\varepsilon u}{2} \right) \mathbb{E}_{N-2} \left[W_{\varepsilon u} \mathbf{1}_{A_{\varepsilon u}} g(\varepsilon u, M(\varepsilon u; \cdot)) \right] du,$$

where $\mathbb{E}_{N-2}$ integrates over ϑ with density p_{N-2} . The same identity holds for real g whenever the right-hand side is finite with $|g|$ in place of g .

Proof. For distinct angles and an ordered pair (a, b) , $a \neq b$, let $\sigma_{ab} \in (0, 2\pi)$ be $\theta_b - \theta_a$ modulo 2π . Call b the successor of a if no θ_j lies on the open counterclockwise arc from θ_a to θ_b . The gaps are in bijection with the ordered pairs (a, b) such that b is the successor of a , the gap being σ_{ab} and the hump the maximum of $|\Lambda_N|$ over the closed arc. By exchangeability each of the $N(N-1)$ ordered pairs contributes equally, so the left side of (9) equals

$$\varepsilon^{-3} N(N-1) \int_{\mathbb{T}^N} p_N(\theta) \mathbf{1}\{\sigma_{12} < \varepsilon\} \mathbf{1}\{2 \text{ is the successor of } 1\} g(\sigma_{12}, M_{12}) d\theta.$$

Change variables $\theta_1 = \varphi$, $\theta_2 = \varphi + \varepsilon u \pmod{2\pi}$, $u \in (0, 1)$, which maps $\mathbb{T} \times (0, 1)$ bijectively onto $\{(\theta_1, \theta_2) : \sigma_{12} < \varepsilon\}$ with $d\theta_1 d\theta_2 = \varepsilon d\varphi du$, and put $\vartheta_j = \theta_{j+2} - \varphi \pmod{2\pi}$, a measure-preserving bijection of $\mathbb{T}^{N-2}$. The Vandermonde product splits into the pair factor $|1 - e^{i\varepsilon u}|^2 = 4 \sin^2(\varepsilon u/2) = \varepsilon^2 u^2 \operatorname{sinc}^2(\varepsilon u/2)$, the coupling $W_{\varepsilon u}(\vartheta)$, and the Vandermonde product of ϑ ; the successor condition becomes $\vartheta \in A_{\varepsilon u}$; and $M_{12} = M(\varepsilon u; \vartheta)$ because $|\Lambda_N(e^{i(\varphi+t)})| = |e^{it} - 1| |e^{it} - e^{i\varepsilon u}| |\Lambda_{N-2}(e^{it})|$. Nothing depends on φ , whose integral gives 2π . Finally

$$\frac{N(N-1)}{(2\pi)^N N!} \prod_{j < k} |e^{i\vartheta_j} - e^{i\vartheta_k}|^2 = \frac{1}{(2\pi)^2} p_{N-2}(\vartheta),$$

and collecting $\varepsilon^{-3} \cdot \varepsilon \cdot \varepsilon^2 \cdot 2\pi/(2\pi)^2 = 1/(2\pi)$ gives (9). The statement for real g follows by splitting $g = g^+ - g^-$. $\square$

3.3. Proof of Theorem 1. We use the generalised dominated convergence theorem: if $|f_\varepsilon| \leq g_\varepsilon$, $f_\varepsilon \rightarrow f$ and $g_\varepsilon \rightarrow g$ almost everywhere, and $\int g_\varepsilon \rightarrow \int g < \infty$ as $\varepsilon \rightarrow 0$, then $\int f_\varepsilon \rightarrow \int f$ (along any sequence $\varepsilon_n \rightarrow 0$, apply Fatou's lemma to $g_{\varepsilon_n} \pm f_{\varepsilon_n} \geq 0$).

Proof of (ii). Apply Lemma C with $g(s, M) = F(s/\varepsilon, M/s^2)$ (admissible by part (i) and the bound below). For $0 < \varepsilon \leq 1$, $u \in (0, 1]$ and $\vartheta \in \mathbb{T}^{N-2}$ put $s = \varepsilon u$ and

$$f_\varepsilon(u, \vartheta) = u^2 \operatorname{sinc}^2\left(\frac{s}{2}\right) W_s(\vartheta) \mathbf{1}_{A_s}(\vartheta) F\left(u, \frac{M(s; \vartheta)}{s^2}\right),$$

so that the left side of (5), before the limit, is $(2\pi)^{-1} \int_0^1 \mathbb{E}_{N-2} f_\varepsilon(u, \cdot) du$.

Domination. On A_s , by the growth condition on F , $\operatorname{sinc} \leq 1$, and (7), (8) (with $k = k_0$),

$$|f_\varepsilon| \leq C u^2 W_s \left(1 + \left(\frac{M}{s^2}\right)^{2k_0}\right) \leq C u^2 \left(|\Lambda_{N-2}(e^{is/2})|^4 + 5^{-2k_0} |\Lambda_{N-2}(e^{is/2})|^{4+2k_0}\right) =: g_\varepsilon(u, \vartheta),$$

and $f_\varepsilon = 0$ off A_s . By rotation invariance $\int_0^1 \mathbb{E}_{N-2} g_\varepsilon du = \frac{C}{3} (\mathcal{M}_{N-2}(2) + 5^{-2k_0} \mathcal{M}_{N-2}(k_0 + 2)) < \infty$ for every ε .

Pointwise limits. Let ϑ be such that no ϑ_j is 0; this excludes a null set, and on its complement $\Lambda_{N-2}(1) \neq 0$. Fix $u \in (0, 1]$ and let $\varepsilon \rightarrow 0$. Then $\vartheta \in A_{\varepsilon u}$ as soon as $\varepsilon u < \min_j \vartheta_j$; $W_{\varepsilon u}(\vartheta) \rightarrow |\Lambda_{N-2}(1)|^4$ and $|\Lambda_{N-2}(e^{i\varepsilon u/2})| \rightarrow |\Lambda_{N-2}(1)| > 0$ by continuity; $M(\varepsilon u; \vartheta)/(\varepsilon u)^2 \rightarrow |\Lambda_{N-2}(1)|/4 \in (0, B_N]$ by Lemma A; and F is continuous at $(u, |\Lambda_{N-2}(1)|/4)$. Hence

$$\begin{aligned} f_\varepsilon &\rightarrow f_0 := u^2 |\Lambda_{N-2}(1)|^4 F(u, |\Lambda_{N-2}(1)|/4), \\ g_\varepsilon &\rightarrow g_0 := C u^2 (|\Lambda_{N-2}(1)|^4 + 5^{-2k_0} |\Lambda_{N-2}(1)|^{4+2k_0}), \end{aligned}$$

and $\int_0^1 \mathbb{E}_{N-2} g_\varepsilon du = \int_0^1 \mathbb{E}_{N-2} g_0 du$ for all ε . Generalised dominated convergence on $(0, 1] \times \mathbb{T}^{N-2}$ with the measure $du \otimes p_{N-2} d\vartheta$ gives (5). $\square$

Proof of (a)–(c). (a) Take $F \equiv 1$: the limit is $\frac{1}{2\pi} \cdot \frac{1}{3} \cdot \mathcal{M}_{N-2}(2) = N^2(N^2 - 1)/(72\pi)$ by (3).

(b) For bounded continuous F on $[0, 1] \times [0, \infty)$, divide (5) by (a):

$$\langle F(s/\varepsilon, M/s^2) \rangle_\varepsilon \rightarrow \int_0^1 3u^2 \tilde{\mathbb{E}}[F(u, X_{N-2}/4)] du,$$

which is the expectation of $F(U^{1/3}, X_{N-2}/4)$ with U independent of X_{N-2} , since $3u^2 du$ on $[0, 1]$ is the law of $U^{1/3}$.

(c) For $k \in (-\frac{5}{2}, 0]$ take $F(u, y) = y^{2k}$ and $k_0 = k$; for $k > 0$, y^{2k} is bounded on $(0, B_N]$ and $k_0 = 0$. For log-moments, $|\log y|^m \leq C_m(1 + y^{-1})$ on $(0, B_N]$, so $k_0 = -\frac{1}{2}$ applies; likewise for $(\log y)^m$ with integer m . Dividing by (a) gives the limits; cumulants are polynomials in moments. For the last statement let $N \geq 3$ and $k \leq -\frac{5}{2}$, so $a := k+2 \leq -\frac{1}{2}$. The density p_{N-2} is continuous and positive near a configuration with $\vartheta_1 = 0$ and the other angles distinct and away from 0, and there $|\Lambda_{N-2}(1)|^{2a} \geq c |2 \sin(\vartheta_1/2)|^{2a}$, which is not integrable near $\vartheta_1 = 0$; hence $\mathcal{M}_{N-2}(a) = \infty$. With $F = y^{2k} \geq 0$, Fatou's lemma applied to the (nonnegative) integrands $f_\varepsilon \rightarrow f_0$ gives $\liminf_\varepsilon \varepsilon^{-3} \mathbb{E} \sum_n \mathbf{1}\{s_n < \varepsilon\} (M_n/s_n^2)^{2k} \geq \frac{1}{6\pi} 4^{-2k} \mathcal{M}_{N-2}(a) = \infty$, while the denominator of $\langle \cdot \rangle_\varepsilon$ stays finite by (a). $\square$

Remark 1 (Finite ε). All ε -dependence in (9) enters through functions of the unordered pair of endpoints $\{e^{\pm is/2}\}$ (after centring the arc at 1), hence through even functions of s . This suggests that the corrections to the limits in Theorem 1 are $O(\varepsilon^2)$; we have not proved it, but the Monte Carlo data of Section 6 are consistent with it. The maximum over a small gap sits asymptotically at its midpoint (Section S1).

4. THE EXACT LAW OF THE LIMITING HUMP

Theorem 2. *Let $n \geq 1$ and let $\xi_1, \dots, \xi_n$ be independent, with ξ_1 on the unit circle with law $\frac{1}{6}|1 - e^{i\alpha}|^4 \frac{d\alpha}{2\pi}$ and, for $2 \leq j \leq n$, ξ_j in the open unit disc $\mathbb{D}$ with density*

$$c_j (1 - |z|^2)^{j-2} |1 - z|^4 \quad (\text{w.r.t. } d^2z), \quad c_j = \frac{j-1}{\pi} \cdot \frac{j(j+1)}{(j+2)(j+3)}.$$

Then $X_n \stackrel{d}{=} \prod_{j=1}^n |1 - \xi_j|$. Moreover:

- (a) $\tilde{\mathbb{E}}X_n^{2k} = \frac{\mathcal{M}_n(k+2)}{\mathcal{M}_n(2)} = \prod_{j=1}^n \frac{\Gamma(j+2)^2 \Gamma(j+4+2k)}{\Gamma(j+4) \Gamma(j+2+k)^2}$ for $k > -\frac{5}{2}$, and $\tilde{\mathbb{E}}X_n^{2k} = \infty$ for $k \leq -\frac{5}{2}$.
- (b) $t \mapsto \log \tilde{\mathbb{E}}e^{t \log X_n} = \log \mathcal{M}_n(2+t/2) - \log \mathcal{M}_n(2)$ is analytic for $\text{Re } t > -5$, and for $r \geq 1$

$$(10) \quad \kappa_r(\log X_n) = \sum_{j=1}^n \left[\psi^{(r-1)}(j+4) - 2^{1-r} \psi^{(r-1)}(j+2) \right];$$

$$\text{in particular } \kappa_1(\log X_n) = H_{n+2} + H_{n+3} - \frac{10}{3}.$$

Proof. By [7, Prop. 2.2], under the Haar measure of $U(n)$, $\det(1 - U) \stackrel{d}{=} \prod_{j=1}^n (1 - \gamma_j)$ with independent γ_j : γ_1 is uniform on the unit circle and, for $j \geq 2$, $\gamma_j = e^{i\omega_j} \sqrt{\beta_j}$ with ω_j uniform and $\beta_j \sim \text{Beta}(1, j-1)$ independent. (There the factors are written $1 + e^{i\omega_j} \sqrt{\beta_j}$, which has the same law since ω_j is uniform.) Passing to $z = \sqrt{\beta} e^{i\omega}$ ($d^2z = \frac{1}{2} d\beta d\omega$), γ_j has density $\frac{j-1}{\pi} (1 - |z|^2)^{j-2}$ on $\mathbb{D}$.

Tilting. Let μ_j be the law of γ_j and $m_j = \int |1 - z|^4 \mu_j(dz)$. For bounded measurable h , by independence,

$$\tilde{\mathbb{E}}h(X_n) = \frac{\mathbb{E}[\prod_j |1 - \gamma_j|^4 h(\prod_j |1 - \gamma_j|)]}{\prod_j m_j} = \int h\left(\prod_j |1 - z_j|\right) \prod_{j=1}^n \nu_j(dz_j),$$

with $\nu_j(dz) = m_j^{-1} |1 - z|^4 \mu_j(dz)$, because the Radon–Nikodym density $\prod_j |1 - z_j|^4 / \prod_j m_j$ is a product of functions of the separate coordinates; so the $\xi_j \sim \nu_j$ are independent. The constants: $m_1 = \int_0^{2\pi} 16 \sin^4(\alpha/2) \frac{d\alpha}{2\pi} = 6$; for $j \geq 2$, with $\beta = |z|^2$ and rotation invariance, $m_j = \mathbb{E}(1 + \beta)^2 + 2\mathbb{E}\beta = 1 + \frac{4}{j} + \frac{2}{j(j+1)} = \frac{(j+2)(j+3)}{j(j+1)}$, since $\mathbb{E}\beta = \frac{1}{j}$ and $\mathbb{E}\beta^2 = \frac{2}{j(j+1)}$. Hence $c_j = \frac{j-1}{\pi m_j}$ as stated. (Equivalently, $m_j = \mathcal{M}_j(2)/\mathcal{M}_{j-1}(2)$.) The same decomposition is the case $\beta = 2$, $\delta = 2$ of the product formula for the circular Jacobi ensemble in terms of deformed Verblunsky coefficients [8, Thm. 3.2 and Prop. 4.3]: their γ_{n-j} has the law of our ξ_j , $1 \leq j \leq n$.

(a) By definition $\tilde{\mathbb{E}}X_n^{2k} = \mathbb{E}_n |\Lambda_n(1)|^{4+2k} / \mathcal{M}_n(2)$, and (2) with $k+2$ in place of k gives the product; it is finite iff $k+2 > -\frac{1}{2}$ (see the proof of Theorem 1(c)).

(b) $\tilde{\mathbb{E}}e^{t \log X_n} = \tilde{\mathbb{E}}X_n^t = \mathcal{M}_n(2+t/2)/\mathcal{M}_n(2)$, and $\log \mathcal{M}_n(2+t/2) = \sum_j [\log \Gamma(j) + \log \Gamma(j+4+t) - 2 \log \Gamma(j+2+t/2)]$ is analytic for $\text{Re } t > -5$. Its r -th derivative at $t = 0$ is $\sum_j [\psi^{(r-1)}(j+4) - 2 \cdot 2^{-r} \psi^{(r-1)}(j+2)]$, which is (10). For $r = 1$, $\psi(j+4) - \psi(j+2) = \frac{1}{j+2} + \frac{1}{j+3}$, and summing gives $(H_{n+2} - H_2) + (H_{n+3} - H_3) = H_{n+2} + H_{n+3} - \frac{10}{3}$. $\square$

Remark 2 (The tilt ladder in the unitary group). The same argument with the tilt $|\Lambda_n(1)|^{2b}$ gives

$$(11) \quad \kappa_r^{(b)}(n) = \sum_{j=1}^n \left[\psi^{(r-1)}(j+2b) - 2^{1-r} \psi^{(r-1)}(j+b) \right].$$

For $b = 0$ and $n = N$ this is $\log |\Lambda_N(1)|$ under Haar measure [22]. For $b = 1$ and $n = N - 1$ it is exact at finite N for the derivative at an eigenvalue: $|\Lambda'_N(e^{i\theta_1})| = \prod_{j \geq 2} |e^{i\theta_1} - e^{i\theta_j}|$, and by (1) the other $N - 1$ angles, given θ_1 , are $|\Lambda_{N-1}(e^{i\theta_1})|^2$ -tilted [20]. For $b = 2$ and $n = N - 2$ it is the small-gap limit of Theorem 1.

5. LARGE N

Corollary 3. *Let $Y_N = (2\pi/N)^2 X_{N-2}/4$, the small-gap limit of $M_J/\tilde{s}_J^2$ (Theorem 1(b) with $\tilde{s} = Ns/2\pi$). As $N \rightarrow \infty$:*

(a) *for every fixed $k > -\frac{5}{2}$,*

$$\mathbb{E} Y_N^{2k} = \left(\frac{\pi}{N} \right)^{4k} \frac{\mathcal{M}_{N-2}(k+2)}{\mathcal{M}_{N-2}(2)} \sim \pi^{4k} \frac{12 G(k+3)^2}{G(2k+5)} N^{k^2}; \quad \text{e.g. } \mathbb{E} Y_N^2 \sim \frac{\pi^4 N}{720};$$

(b) $\mathbb{E} \log Y_N = H_N + H_{N+1} - \frac{10}{3} - \log 4 + 2 \log \frac{2\pi}{N} = 2 \log \pi + 2\gamma - \frac{10}{3} + O(N^{-1})$, and $2 \log \pi + 2\gamma - \frac{10}{3} = 0.110558\dots$;

(c) $\text{Var} \log Y_N = \frac{1}{2} \log N + c_2 + O(N^{-1})$, with $c_2 = \frac{65}{18} + \frac{\gamma}{2} - \frac{\pi^2}{2} = -1.035083\dots$;

(d) $\kappa_3(\log Y_N) < 0$ for every $N \geq 3$, it decreases strictly with N , and

$$\begin{aligned} \kappa_3(\log Y_N) \rightarrow & -2[\zeta(2) - H_4^{(2)} - 4(\zeta(3) - H_4^{(3)})] \\ & + \frac{1}{2}[\zeta(2) - H_2^{(2)} - 2(\zeta(3) - H_2^{(3)})] = -0.127077\dots; \end{aligned}$$

more generally, for $r \geq 3$, $\kappa_r(\log Y_N) \rightarrow (-1)^r (r-1)! [A_4^{(r)} - 2^{1-r} A_2^{(r)}]$ with $A_a^{(r)} = \zeta(r-1) - H_a^{(r-1)} - a(\zeta(r) - H_a^{(r)})$;

(e) $(\log Y_N - \mathbb{E} \log Y_N) / \sqrt{\frac{1}{2} \log N} \Rightarrow \mathcal{N}(0, 1)$.

The typical close-pair hump in unfolded units, $M/\tilde{s}^2$, stays of order one, but its logarithm spreads like $\frac{1}{2} \log N$; the linear growth of $\mathbb{E} Y_N^2$ is carried by that log-normal tail. The $O(N^{-1})$ terms in (c) are large at moderate N : the exact variance is 0.2987 at $N = 10$ and 0.6340 at $N = 24$, against 0.1162 and 0.5539 from $\frac{1}{2} \log N + c_2$. Finite- N comparisons should use (10).

Proof (the two auxiliary lemmas are proved in Section S2). (a) By Theorem 1(c) and scaling, $\mathbb{E} Y_N^{2k} = (2\pi/N)^{4k} 4^{-2k} \tilde{\mathbb{E}} X_{N-2}^{2k}$. From $\prod_{j=1}^n \Gamma(j+c) = G(n+1+c)/G(1+c)$ and the classical expansion of $\log G$ [3], $\mathcal{M}_n(a) \sim \frac{G(1+a)^2}{G(1+2a)} n^{a^2}$ for fixed $a > -\frac{1}{2}$ [22]. With $a = k+2$ and $a = 2$, $G(3) = 1$, $G(5) = 12$, this gives $\mathcal{M}_{N-2}(k+2)/\mathcal{M}_{N-2}(2) \sim 12 G(k+3)^2 G(2k+5)^{-1} N^{(k+2)^2-4}$; multiply by $(\pi/N)^{4k}$. For $k = 1$, $12 G(4)^2/G(7) = 12 \cdot 4/34560 = 1/720$.

(b) $\kappa_1(\log X_{N-2}) = H_N + H_{N+1} - \frac{10}{3}$ by Theorem 2; add $\log \frac{1}{4} + 2 \log \frac{2\pi}{N}$ and use $H_m = \log m + \gamma + O(m^{-1})$.

(c) By (10), $\kappa_2 = S_4 - \frac{1}{2} S_2$ with $S_a = \sum_{j \leq n} \psi'(j+a) = \log n + \gamma - H_a - a(\zeta(2) - H_a^{(2)}) + 1 + O(n^{-1})$, $n = N - 2$, by (S1). With $H_4 = \frac{25}{12}$, $H_4^{(2)} = \frac{205}{144}$, $H_2 = \frac{3}{2}$, $H_2^{(2)} = \frac{5}{4}$ this is $\frac{1}{2} \log N + \frac{65}{18} + \frac{\gamma}{2} - \frac{\pi^2}{2} + O(N^{-1})$. Scaling does not change the variance.

(d) The summands $\tau_j = \psi''(j+4) - \frac{1}{4}\psi''(j+2)$ of $\kappa_3(\log X_n)$ are all negative: $\tau_1 = -0.010261\dots$, and for $j \geq 2$ the bounds $x^{-2} \leq |\psi''(x)| \leq x^{-2} + 2x^{-3}$ reduce the claim to $4^{1/3}(j+2) > j+4$, true for $j \geq 2$. Hence $\kappa_3(\log X_n)$ is negative and strictly decreasing in n , and its limit is $\sum_{j \geq 1} \tau_j$, evaluated with $\sum_{j \geq 1} \psi^{(r-1)}(j+a) = (-1)^r(r-1)!A_a^{(r)}$ ($r \geq 3$).

(e) By (c) and (d), $\kappa_2 \rightarrow \infty$ while κ_r stays bounded for each $r \geq 3$, so the cumulants of the standardised variable converge to those of $\mathcal{N}(0,1)$; the moments converge, and since the normal law is determined by its moments, the standardised variable converges in law. Replace $\sqrt{\kappa_2}$ by $\sqrt{\frac{1}{2} \log N}$ (Slutsky). $\square$

6. MONTE CARLO CHECK

Haar unitaries were generated by the QR decomposition of complex Ginibre matrices with the phases of the diagonal of R removed [25]; for every gap with unfolded length $\tilde{s} < 0.4$ the maximum of $|\Lambda_N|$ was located on a 41-point grid across the gap with parabolic refinement. Table 1 compares the log-statistics with Theorems 1 and 2: at $\tilde{s} < 0.1$ every deviation of the log-mean is below 2σ , and the mean hump $\mathbb{E}[M/s^2]$ and the rate of small gaps also agree within 2σ (Table S1). Further checks (Section S4): the correlation of $\tilde{s}$ with $\log(M/s^2)$ falls to 0.000 at $\tilde{s} < 0.1$; independent runs give $\kappa_3 = -0.0639$ and -0.0863 at $N = 10$ and 16 , against -0.0688 and -0.0872 ; sampling the product of Theorem 2 directly reproduces its moments; and at finite ε the excess of $\mathbb{E}[M/s^2]$ over its limit scales like ε^2 , as Remark 1 suggests. No zeta data enter this section.

N	$\mathbb{E} \log(M/s^2)$			$\text{Var} \log(M/s^2)$			κ_3
	Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$	Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$	Thm
6	0.3232	0.3276(22)	0.3229(62)	0.1486	0.1447	0.1448	-0.0439
10	1.2292	1.2323(29)	1.2330(81)	0.2987	0.2963	0.2834	-0.0688
16	2.1007	2.0998(34)	2.0822(95)	0.4691	0.4664	0.4781	-0.0872
24	2.8723	2.8827(39)	2.8804(110)	0.6340	0.6306	0.6326	-0.0990

TABLE 1. CUE Monte Carlo against Theorems 1 and 2 (angle units; statistical errors in the last digits; $6 \cdot 10^5$, $4 \cdot 10^5$, $3 \cdot 10^5$, $2 \cdot 10^5$ matrices for $N = 6, 10, 16, 24$).

7. THE ZETA SIDE: A TILT LADDER AND A CONJECTURE

Observables. Let $L = \log(t/2\pi)$, and for consecutive zeros let $m_n = (\gamma_n + \gamma_{n+1})/2$, $L_n = \log(m_n/2\pi)$, $\tilde{\delta}_n = \delta_n L_n/2\pi$ and

$$Y_n := M_n/\tilde{\delta}_n^2.$$

For a close pair, $Z(t) \approx \frac{1}{2}Z''(m_n)(t - \gamma_n)(t - \gamma_{n+1})$ on the gap, so $M_n = \frac{1}{8}|Z''(m_n)|\delta_n^2(1 + O((\delta_n L)^2))$ and Y_n is the zeta analogue of $M/\tilde{s}^2$. The three rungs of the *tilt ladder* are

- $b = 0$: $x = \log |Z(t)|$ at random t (Keating–Snaith [22]);
- $b = 1$: $x = \log (|Z'(\gamma_n)| 2\pi/L)$ at the zeros (Hughes–Keating–O’Connell [20]);
- $b = 2$: $x = \log Y_n$ on close pairs, $\tilde{\delta}_n < \varepsilon$ (this note).

The random-matrix baseline of rung b is $\kappa_r^{\text{CUE},b}(N) := \kappa_r^{(b)}(N-b)$ of (11), evaluated at $N = L$ by analytic continuation in N : the sums in (11) are replaced by

$$\begin{aligned} \sum_{j=1}^n \psi^{(r-1)}(j+a) &= (n+a) \psi^{(r-1)}(n+a+1) - a \psi^{(r-1)}(a+1) \\ &\quad + (r-1) [\psi^{(r-2)}(n+a+1) - \psi^{(r-2)}(a+1)] \end{aligned}$$

for $r \geq 2$ (for $r = 1$ the bracket is replaced by $-n$), an identity for integer $n \geq 1$ (induction on n) whose right side is analytic in $n > -a - 1$.

Arithmetic cumulants. The Keating–Snaith conjecture [22] multiplies the moments of $|\zeta|$ by

$$a_k = \prod_p \left(1 - \frac{1}{p}\right)^{k^2} \sum_{m \geq 0} \left(\frac{\Gamma(m+k)}{m! \Gamma(k)}\right)^2 p^{-m},$$

and the Hughes–Keating–O’Connell conjecture for $\zeta'(\rho)$ [20] carries the same factor. Writing $\log a_{t/2} = \sum_r \kappa_r^{\text{arit}} t^r / r!$ and expanding ${}_2F_1(k, k; 1; x) = 1 + k^2 \text{Li}_2(x) + 2k^3 S_{1,2}(x) + O(k^4)$, with $S_{1,2}(x) = \sum_{m \geq 2} H_{m-1} x^m / m^2$, gives

$$\begin{aligned} \kappa_1^{\text{arit}} &= 0, & \kappa_2^{\text{arit}} &= \frac{1}{2} \sum_p \left[\log\left(1 - \frac{1}{p}\right) + \text{Li}_2\left(\frac{1}{p}\right) \right] = -0.088124, \\ \kappa_3^{\text{arit}} &= \frac{3}{2} \sum_p S_{1,2}\left(\frac{1}{p}\right) = +0.233653 \end{aligned}$$

(primes up to 3×10^6 ; the tails are below 10^{-8}).

Conjecture 4 (Close pairs; as formulated before the tests). *Let $T \rightarrow \infty$ and then $\varepsilon \rightarrow 0$, for zeros in $[T, 2T]$ with $\tilde{\delta}_n < \varepsilon$.*

- (i) $(\tilde{\delta}_n/\varepsilon, Y_n)$ are asymptotically independent, and $\tilde{\delta}_n/\varepsilon \xrightarrow{d} U^{1/3}$.
- (ii) $\mathbb{E}[Y_n^{2k} \mid \tilde{\delta}_n < \varepsilon] \sim a_k \mathbb{E}Y_L^{2k} \sim a_k \pi^{4k} 12 G(k+3)^2 G(2k+5)^{-1} L^{k^2}$, where Y_L is Y_N of Corollary 3 at $N = L$; in particular $\mathbb{E}[Y_n^2 \mid \tilde{\delta}_n < \varepsilon] \sim \pi^4 L / 720$ ($a_1 = 1$).
- (iii) $\kappa_r(\log Y_n \mid \tilde{\delta}_n < \varepsilon) = \kappa_r^{\text{CUE},2}(L) + \kappa_r^{\text{arit}} + o(1)$.

The heuristic is the hybrid Euler–Hadamard product of Gonek, Hughes and Keating [18], $\zeta \approx P_X Z_X$: conditioning on a close pair acts on Z_X as in Theorem 1, while P_X is not tilted at leading order, as in the treatment of $\zeta'(\rho)$ by Bui, Gonek and Milinovich [9]; the moments factor, and X cancels between $\mathbb{E}|P_X|^{2k} \sim a_k (e^\gamma \log X)^{k^2}$ and the tilted CUE of dimension $L/(e^\gamma \log X)$. Its weak points, recorded before the tests (Section S5), are the independence of P_X from the conditioning — the prime waves modulate the local density of zeros at relative order $1/L$ [23, 19, 2], and at $b = 2$ the arithmetic term is not a small correction — finite L , the order of limits, and the passage from (ii) to (iii). At $L = 10$ every model with an arithmetic factor predicts a positive κ_3 on $b = 2$, whereas the pure tilted CUE gives -0.069 (Table S2). The pre-test headline was therefore a sign: *close-pair log-humps of zeta are skewed to the right*, with skewness $\kappa_3/\kappa_2^{3/2} \approx +1.7$ at $L = 10$ under a_k .

8. PRE-REGISTERED TESTS ON THE RIEMANN ZEROS

Protocol and data. Each test was frozen — predictions, thresholds, analysis code — in a pre-registration file that was committed with its sha256 stamp to the author’s private working repository, and pushed, before the measurement (Tests 200-A, B, C on 26–27 September 2026, Test 201 on 28 September 2026). The files and stamps are in the public repository; the commit hashes (Section S8) refer to the private one, whose history can be shown to referees on request. Test 200-B was designed after 200-A was sealed, and 200-C after 200-B. Before each measurement no distribution summary of the tested observable had been computed on the data used (the one earlier look, a Kolmogorov–Smirnov test of $\log|Z(t)|$ at low height, is described in Section S6). Tests 200-A/B use A. M. Odlyzko’s first 2×10^6 zeros [27, 26] in three windows W1–W3 ($\bar{L} \approx 9.3, 10.6, 11.65$); Test 200-C uses four LMFDB files of 1.5×10^6 consecutive zeros computed by Platt [24, 28], at $\bar{L} = 14.19, 16.58, 18.88, 22.31$ (C1–C4). Z and Z' are computed by a vectorised Riemann–Siegel formula checked against `mpmath`. On rung $b = 0$ we use 10^6 uniform random t per window, on $b = 1$ all zeros (10^6 random zeros in C4), on $b = 2$ all pairs with $\tilde{\delta}_n < 0.2$ (1 143 to 12 143 per window). Statistics are unbiased k -statistics with block-jackknife errors calibrated on synthetic data. The *arithmetic shift* of rung b is $s_b := k_3 - \kappa_3^{\text{CUE},b}(\bar{L})$ (similarly for k_2); on $b = 2$ a finite- ε CUE correction is also subtracted, and $\bar{D}_r$ denotes the resulting $b = 2$ shift of k_r pooled over W1–W3. The full protocol and all numbers are in Section S6.

window	L	$b = 0$	$b = 1$	$b = 2$	s_1/s_0	s_2/s_1
W1–W3	9.3–11.7	+0.2828(93)	+0.1499(9)	+0.0859(30)	0.53	0.57
C1	14.19	+0.281(16)	+0.1633(13)	+0.1053(77)	0.58	0.64
C2	16.58	+0.289(20)	+0.1744(12)	+0.1183(60)	0.60	0.68
C3	18.88	+0.275(17)	+0.1787(15)	+0.1242(69)	0.65	0.70
C4	22.31	+0.283(19)	+0.1839(19)	+0.1466(91)	0.65	0.80
a_k		+0.2337	+0.2337	+0.2337	1	1

TABLE 2. Arithmetic shift of κ_3 (observed minus tilted CUE at $N = \bar{L}$; on $b = 2$ after the finite- ε correction) by rung and height; W1–W3 pooled. The variance shifts lie between -0.073 and -0.115 on every rung and at every height, around $\kappa_2^{\text{arit}} = -0.088$; they do not decay along the ladder.

Results. *200-A (lower rungs).* Among parameter-free models (CUE, a_k , hybrid Euler products, Gaussian surface) the twelve numbers (k_2, k_3 on two rungs in three windows) select a_k decisively ($\chi^2 = 28.2$ for 12 points; CUE 315.2, hybrids 872–1243, Gaussian 3689.7). Post hoc, the variance shift is close to κ_2^{arit} on both rungs, but the κ_3 shift is not: $+0.283$ at $b = 0$ and $+0.150$ at $b = 1$, against $+0.234$ (Table 2). Before 200-B we registered, besides a_k , two extrapolations of the two lower rungs (not theories): geometric decay, $s_2 = s_1^2/s_0$, predicting $\bar{D}_3 = +0.0794 \pm 0.0028$ on $b = 2$, and linear decay.

200-B (close pairs, blind). With 14 168 close pairs ($\tilde{\delta} < 0.2$, $L \approx 9$ –12) the arithmetic signature is present on $b = 2$ ($\bar{D}_3 = +0.0859 \pm 0.0030$, 29σ) and the skewness is positive (corrected $\kappa_3 = +0.0143 \pm 0.0030$, 4.8σ): the headline prediction held in sign but not in size (skewness about $+0.15$, not $+1.7$). The a_k model holds for the variance and fails for κ_3 ; the shape of the ladder is geometric, decisively ($\chi^2 = 0.36$ against 24.0 for an extinguished and

200.7 for a constant shift), and the blind geometric prediction differs from the measurement by 1.6σ . The variance shift on $b = 2$, -0.0988 ± 0.0036 , misses the geometric and linear extrapolations of the lower rungs (-0.064) by about 9σ .

200-C (height, blind). Before 200-C, a post hoc fit of the W1–W3 values to $s_b(L) = a_k + c_b L^{-\nu}$ gave $\nu = 0.48 \pm 0.12$, and three classes were registered for the new windows: this relaxation ($H_{S\nu}$), the same with $\nu = 1$ (H_{S1}), and constant shifts (H_C). On C1–C4 the $b = 1$ and $b = 2$ shifts rise towards a_k , while $b = 0$ stays near 0.28 within its larger errors (Table 2); the class is $H_{S\nu}$ ($\chi^2 = 10.3$ for 12 points; H_{S1} 30.1, H_C 166.2). Fitted on the new windows alone with asymptote a_k , the exponent is $\hat{\nu} = 0.749 \pm 0.170$; over all seven windows a free common asymptote is 0.277 ± 0.041 , within 1σ of κ_3^{arit} . The observed $b = 1$ shifts lie 0.005–0.01 above the blind predictions (faster relaxation).

201 (phase lock). For single zeros Landau’s formula [23, 19] gives $\mathbb{E}[\cos(m\gamma_n \log p)] \approx -\log p/(p^{m/2}L)$; the measured lock agrees to about four decimals (ratio 1.0000 ± 0.00004 over seven windows and $p \leq 13$). Midpoints of close pairs ($\tilde{\delta}_n < 0.2$) lock $R_2 = 3.874 \pm 0.023$ times as strongly, the class 4 registered from a local-GUE heuristic (pairs selected with the mean density have midpoints weighted by $n(t)^4$), with the classes 1 and 2 excluded; R_2 decreases with ε as the heuristic expects. The lock has the predicted size, so it does not by itself explain the decay of the κ_3 shift along the ladder.

9. FINITE HEIGHT FROM THE RATIOS CONJECTURE

The tests left the flat $b = 0$ shift (≈ 0.28 against a_k ’s 0.234) unexplained. Its limit is settled by a theorem: Fazzari and Gerspach [16] prove, under the Riemann Hypothesis, a triple-correlation hypothesis and a twisted pair-correlation conjecture, that $\frac{1}{T} \int_T^{2T} (\log |\zeta(\frac{1}{2} + it)|)^3 dt = -\pi^2/4 + c_P + O(1/\log T)$ with $c_P = \frac{3}{4} \sum_{p,m \geq 2} \frac{1}{mp^m} \sum_{k+\ell=m} \frac{1}{k\ell}$. Since $\sum_{k+\ell=m} 1/(k\ell) = 2H_{m-1}/m$, this is $c_P = \frac{3}{2} \sum_p S_{1,2}(1/p) = \kappa_3^{\text{arit}}$ exactly: $-\pi^2/4$ is the $N \rightarrow \infty$ limit of the CUE value and c_P the a_k shift. What the theorem leaves open is the finite-height term, and this is what the data at $L = 9$ –22 measure.

We computed it from the ratios conjecture of Conrey, Farmer and Zirnbauer [10]. Writing $\log \zeta(s) = -\int_0^\infty \frac{\zeta'}{\zeta}(s+a) da$, the second and third cumulants of $\log |\zeta(\frac{1}{2} + it)|$ are integrals over the shifts of the two- and three-point averages of ζ'/ζ . The two-point average is that of Conrey and Snaith [12]; the three-point average, derived in Appendix A, consists of an identity term, which gives exactly c_P , and two swap terms carrying the factor $(t/2\pi)^{-x}$ and an arithmetic kernel, which tend to $-\pi^2/4$ as $L \rightarrow \infty$ and supply the finite-height correction. The resulting predictions contain no free parameter; they were computed and committed before being compared with the data (the comparison is therefore not part of the pre-registered tests, but nothing in it was adjusted). Table 3 shows the outcome: both cumulants agree at all seven heights, $\chi^2 = 6.4$ (κ_2) and 2.5 (κ_3) for 7 points, whereas the limit $-\pi^2/4 + c_P$ alone gives $\chi^2 = 274$ and the a_k model (tilted CUE at $N = L$ plus κ_3^{arit}) gives $\chi^2 = 58$. The finite-height correction of ζ is not that of the CUE at $N = L$; the difference comes from the arithmetic kernel of the swap terms, i.e. from the interaction of primes and zeros. We have not found this computation in the literature: the ratios conjecture has been used for the lower-order terms of n -point correlations of zeros [11], and the leading constant of the third moment was obtained by a different route in [16], without numerics.

The same averages reach the second rung. Weighting by the density of zeros, $\sum_\gamma f(\gamma) = \int f(t) \frac{1}{2\pi} [L + 2 \operatorname{Re} \frac{\zeta'}{\zeta}(\frac{1}{2} + it)] dt$, turns averages over zeros into averages over t with one more

factor ζ'/ζ , so that the variance of $\log |\zeta'(\rho)|$ needs only the two- and three-point averages already used (Appendix A; in the CUE analogue the same computation reproduces the exact Palm values to six decimals). The data of this rung have error bars five to eight times smaller than at $b = 0$. Committed before comparison, the prediction gives $\chi^2 = 10.9$ for 7 points (10.7 when it is averaged over the range of L within each window; largest deviation 2.9σ , i.e. 0.0015, at $L = 16.58$), whereas the a_k model gives $\chi^2 = 1780$: the +0.015 excess of the $b = 1$ variance over a_k seen in Test 200-A is again a finite-height effect.

The third cumulant over zeros needs the four-point averages $\mathbb{E}[XXX\bar{X}]$ and $\mathbb{E}[XX\bar{X}\bar{X}]$, including a term in which both unconjugated shifts are swapped (Appendix A). We checked the computation in three independent ways before comparing with the data: the CUE analogue reproduces the exact Palm value $\sum_{j < N} [\psi''(j+2) - \frac{1}{4}\psi''(j+1)]$ to 10^{-7} ; every arithmetic factor was checked prime by prime against numerical integration over the phase; and the connected three-point cumulants over zeros that enter the integrand were computed directly from $1.5 \cdot 10^6$ zeros at $L = 14.19$, agreeing at all fourteen tested shift triples (largest pull 1.1) while the CUE is off by 9–26%. The committed prediction gives $\chi^2 = 11.6$ for 7 points (largest deviation 2.2σ); an independent audit after the comparison found one error, a tail term too large by a factor of about ten in one arithmetic sum, whose correction moves the predictions by $3\text{--}5 \cdot 10^{-4}$ and gives $\chi^2 = 10.9$. The a_k model gives $\chi^2 = 16\,826$ and even has the wrong sign: at these heights the third cumulant of $\log |\zeta'(\rho)|$ is negative, near -0.047 , while tilted CUE plus κ_3^{arit} is positive. The halving of the κ_3 shift from $b = 0$ to $b = 1$ that opened this study is therefore a finite-height effect of the interaction between primes and zeros, and the ratios conjecture accounts for it quantitatively.

L	$b = 0: \log \zeta(\frac{1}{2} + it) $		$b = 1: \log \zeta'(\rho) $	
	κ_2	κ_3	κ_2	κ_3
9.35	1.8191(45) / 1.8249	−2.089(18) / −2.095	0.4672(22) / 0.4691	−0.0465(27) / −0.0488
10.59	1.8899(44) / 1.8858	−2.124(20) / −2.109	0.5200(14) / 0.5190	−0.0519(21) / −0.0492
11.66	1.9348(39) / 1.9330	−2.124(13) / −2.118	0.5586(11) / 0.5584	−0.0509(10) / −0.0490
14.19	2.0333(40) / 2.0300	−2.133(16) / −2.136	0.64107(55) / 0.64099	−0.0487(13) / −0.0480
16.58	2.1051(42) / 2.1068	−2.133(20) / −2.149	0.70628(54) / 0.70783	−0.0440(12) / −0.0467
18.88	2.1729(40) / 2.1714	−2.153(17) / −2.158	0.76478(61) / 0.76496	−0.0445(15) / −0.0454
22.31	2.2471(43) / 2.2542	−2.151(19) / −2.169	0.83829(82) / 0.83925	−0.0446(19) / −0.0437
χ_7^2	6.4	2.5	10.9	11.6

TABLE 3. Observed / ratios-conjecture values of the second and third cumulants of $\log |\zeta(\frac{1}{2} + it)|$ at random t (rung $b = 0$) and of $\log |\zeta'(\rho)|$ over zeros (rung $b = 1$). The predictions were committed before comparison and have no free parameters (Appendix A); for $b = 1$, κ_3 the committed values are shown, and the post-audit correction gives $\chi^2 = 10.9$. As $L \rightarrow \infty$, κ_3 at $b = 0$ tends to $-\pi^2/4 + c_P = -2.2337$ (the prediction is -2.2072 at $L = 60$ and -2.2186 at $L = 120$).

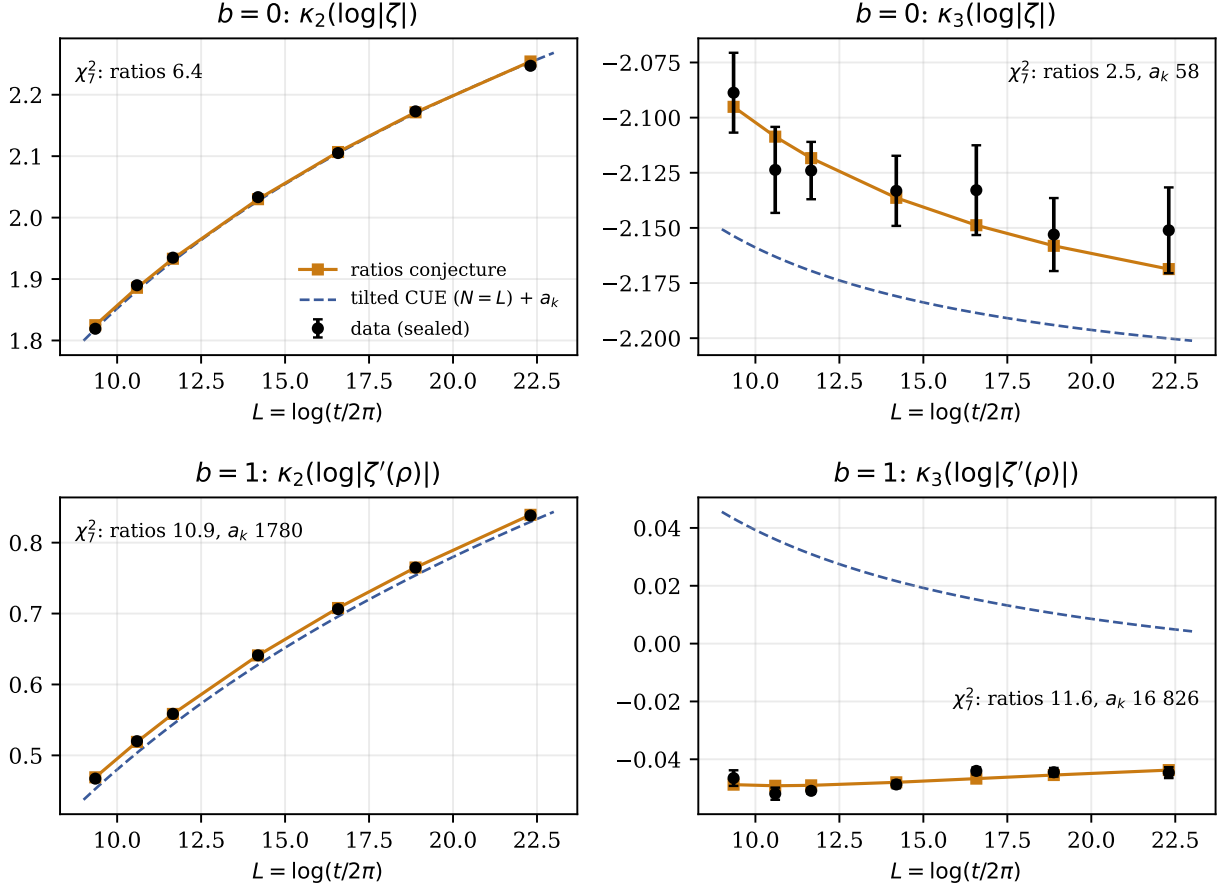


FIGURE 1. The first two rungs of the ladder at finite height. Points: sealed data from Tests 200-A and 200-C (seven windows). Squares: ratios-conjecture predictions, committed before comparison, no free parameters. Dashed: tilted CUE at $N = L$ plus the Keating–Snaith arithmetic cumulants, the model that is correct as $L \rightarrow \infty$.

10. READING AND OPEN PROBLEMS

The “halving at each conditioned zero” seen at $L \approx 11$ is a finite-height effect: as the height grows the ladder flattens, the ratios rising from 0.53/0.57 to 0.65/0.80 (Table 2), the differences between rungs close like $L^{-\nu}$ with $\nu \approx 0.45$ –0.75, and the three rungs approach a common value, as in the asymptotic picture of Keating–Snaith and Hughes–Keating–O’Connell. For $b = 0$ and $b = 1$ the finite-height values are now derived from the ratios conjecture: the plateau of the $b = 0$ shift at ≈ 0.28 is an artefact of subtracting the CUE at $N = L$, whose own $1/N$ term ($+\frac{3}{4}/N$ for κ_3) differs from that of ζ . For $b = 2$ the geometric decay of 200-B and the relaxation law of 200-C remain extrapolations: each passed a blind test on unseen data, but neither is derived. Open problems:

- (1) *Finite- ε corrections.* Prove that the corrections in Theorem 1 are $O(\varepsilon^2)$ and compute the leading coefficient.
- (2) *Close pairs of zeta.* Extend the ratios-conjecture computation of Appendix A to $b = 2$, which needs the two-point density of zeros and hence four- and five-point averages,

and derive the rung dependence of the κ_3 shift and its relaxation; decide whether the phase lock of Test 201 is the mechanism; prove Conjecture 4(ii) for small k under a hypothesis, in the manner of [9] for $\zeta'(\rho)$.

- (3) *Rigour*. Make the $b = 0$ and $b = 1$ computations rigorous under explicit hypotheses, as [16] did for the limit at $b = 0$.
- (4) *Phase lock of close pairs*. Derive the factor 4 of Test 201 and its ε -dependence beyond the local-GUE heuristic.
- (5) *All gaps*. Theorem 1 is the small-gap end of the open problem of [1]: the mean of M_n^2 conditioned on a gap of any size.

APPENDIX A. CUMULANTS OF $\log|\zeta|$ AT FINITE HEIGHT FROM THE RATIOS CONJECTURE

Let $L = \log(t/2\pi)$, $X(\alpha) = \frac{\zeta'}{\zeta}(\frac{1}{2} + \alpha + it)$ and $\bar{X}(\beta) = \frac{\zeta'}{\zeta}(\frac{1}{2} + \beta - it)$ for real shifts, and let $\mathbb{E}$ denote the average over t in a window at height L . Since $\log \zeta(s) = -\int_0^\infty \frac{\zeta'}{\zeta}(s+a) da$ and averages of products of unconjugated factors vanish,

$$(12) \quad \kappa_2 = \frac{1}{2} \iint_{\mathbb{R}_+^2} \mathbb{E}[X(a)\bar{X}(b)] da db, \quad \kappa_3 = -\frac{3}{4} \iiint_{\mathbb{R}_+^3} \mathbb{E}[X(a)X(b)\bar{X}(c)] da db dc$$

for the cumulants of $\log|\zeta(\frac{1}{2} + it)|$ (the mean is 0). We evaluate the averages with the ratios conjecture [10]: the recipe replaces $\mathbb{E} \prod \zeta(s + \alpha_k) \prod \zeta(1 - s + \beta_l) / (\cdots)$ by a sum over swaps $\alpha_S \leftrightarrow -\beta_T$ ($|S| = |T|$), each weighted by $(t/2\pi)^{-\sum_S \alpha - \sum_T \beta}$ and by a product of $\zeta(1 + \cdot)$ factors and an absolutely convergent Euler product A ; the averages of ζ'/ζ follow by differentiating and setting the denominator shifts equal to the numerator shifts. Put $F(x) = e^{-Lx} \zeta(1+x) \zeta(1-x) A(x)$ with

$$A(x) = \prod_p \frac{(1 - p^{-1-x})(1 - 2p^{-1} + p^{-1-x})}{(1 - p^{-1})^2}.$$

Two points [12]. $\mathbb{E}[X(\alpha)\bar{X}(\beta)] = G(\alpha + \beta)$ with $G(x) = \sum_p \frac{\log^2 p}{p^{1+x-1}} + F(x)$, so $\kappa_2 = \frac{1}{2} \int_0^\infty x G(x) dx$.

Three points. With $x_i = \alpha_i + \beta$,

$$(13) \quad \mathbb{E}[X(\alpha_1)X(\alpha_2)\bar{X}(\beta)] = D + F(x_1) K(\alpha_2 - \alpha_1, x_1) + F(x_2) K(\alpha_1 - \alpha_2, x_2),$$

$$D = - \sum_p \log^3 p \frac{y_1 y_2}{(1 - y_1)(1 - y_2)}, \quad y_i = p^{-1-x_i},$$

$$K(u, x) = - \sum_p \log p \frac{q}{1 - q} \frac{(1 - p^{-1})(1 - p^{-x})}{1 - 2p^{-1} + p^{-1-x}}, \quad q = p^{-1-u}.$$

The identity term D is the average under independent uniform phases p^{-it} . The swap term $\alpha_1 \leftrightarrow -\beta$ vanishes to second order at the diagonal through the factors $1/\zeta(1 - \beta + \delta)$ and $1/\zeta(1 - \alpha_1 + \gamma_1)$, which absorb ∂_β and ∂_{α_1} ; ∂_{α_2} acts on the rest. For each prime the local average of the swapped Euler factor has the closed form $1 + (1 - p^{-\alpha_1 - \delta}) \left[\frac{(1 - p^{-1 + \alpha_1 - \gamma_1})(1 - p^{-1 + \alpha_1 - \gamma_2})}{(1 - p^{-1 + \alpha_1 + \beta})(1 - p^{-1 + \alpha_1 - \alpha_2})} - 1 \right]$, and $\frac{\zeta'}{\zeta}(1 + \alpha_2 - \alpha_1) - \frac{\zeta'}{\zeta}(1 + \alpha_2 + \beta) + \partial_{\alpha_2} \log A_{\text{swap}}$ collapses to K (we checked both steps numerically, prime by prime, to 10^{-10}). For $u < 0$ the kernel is continued as $K = \frac{\zeta'}{\zeta}(1 + u) + P(1 + u + x) + Q_2 - C$, with $P(s) = \sum_p \log p p^{-s}$ and two sums Q_2, C that converge for $u > -1$.

The integral. Integrating D term by term gives $\frac{3}{4} \sum_p \sum_{m_1, m_2 \geq 1} \frac{p^{-m_1-m_2}}{m_1 m_2 (m_1+m_2)} = c_P$ exactly. The swap terms are not separately absolutely integrable near the origin: splitting them and taking a principal value at $\alpha_1 = \alpha_2$ loses exactly $-\pi^2/4$. The correct form uses the combined kernel $\Phi(x_1, x_2) = F(x_1)K(x_2 - x_1, x_1) + F(x_2)K(x_1 - x_2, x_2)$, finite on the diagonal; with $x_1 = a + c$, $x_2 = b + c$ the c -integral gives $\min(x_1, x_2)$ and

$$\kappa_3 = c_P - \frac{3}{2} \int_0^\infty x_1 dx_1 \int_0^\infty \Phi(x_1, x_1 + w) dw.$$

With $\zeta(1+x) \mapsto (1-e^{-x})^{-1}$, $A \mapsto 1$ and $L \mapsto N$ the same formula reproduces the exact CUE values $\kappa_3(N) = \frac{3}{4} \sum_{j \leq N} \psi''(j)$ for $N = 1, 2, 5, 10$ to five decimals. Numerical details (prime cut-off, quadrature, cuts) and their sensitivities, all below 10^{-3} , are in Section S7.

Averages over zeros. From the Hadamard product, $\sum_\gamma \frac{c}{c^2+(t-\gamma)^2} = \operatorname{Re} \frac{\zeta'}{\zeta}(\frac{1}{2} + c + it) + \frac{L}{2} + O(1/t)$, so as $c \rightarrow 0^+$ the zeros have density $\frac{1}{2\pi}[L + 2 \operatorname{Re} X(0)]$ and $\mathbb{E}_\rho[M] = \mathbb{E}[M] + \frac{1}{L}(\mathbb{E}[MX(0)] + \mathbb{E}[M\bar{X}(0)])$ (for the CUE this is exact, with N for L). With $\log \zeta'(\rho) = -\int_0^\infty [\frac{\zeta'}{\zeta}(\rho+a) - \mathbf{1}_{a<1}/a] da$, the variance of $\log |\zeta'(\rho)|$ is $\frac{1}{2}[\operatorname{Cov}(Y, \bar{Y}) + \operatorname{Cov}(Y, Y)]$, where $\mathbb{E}_\rho[X(a)] = G(a)/L$, $\mathbb{E}_\rho[X(a)\bar{X}(b)] = G(a+b) + [T(a, 0; b) + T(b, 0; a)]/L$ and $\mathbb{E}_\rho[X(a)X(b)] = T(a, b; 0)/L$, with $T(\alpha_1, \alpha_2; \beta)$ the three-point average (13). The integrand is symmetric, and near $a = 0$ terms of order $1/a$ cancel. In the CUE analogue this reproduces $\sum_{j \leq N} [\psi'(j+2) - \frac{1}{2}\psi'(j+1)]$ to six decimals. As an independent check, the averages $\mathbb{E}_\rho[X(a)\bar{X}(b)]$ entering the integrand were computed directly from $8 \cdot 10^5$ zeros at $L = 14.19$ (where $\operatorname{Re} X_\rho$ is an exact sum over zeros): for a, b between $0.1/L$ and $6/L$ all fifteen cells agree with the formula within their statistical errors, whereas the CUE at $N = L$ is off by about 25%.

Four points: the third cumulant over zeros. Here $\mathbb{E}_\rho[X(a)X(b)\bar{X}(c)] = T(a, b; c) + [Q(a, b, 0; c) + Q_2(a, b; c, 0)]/L$ and $\mathbb{E}_\rho[X(a)X(b)X(c)] = Q(a, b, c; 0)/L$, with the four-point averages

$$\begin{aligned} Q(\alpha_1, \alpha_2, \alpha_3; \beta) &= D_4 + \sum_i F(x_i) [f_{jk}^{(i)} + K(\alpha_j - \alpha_i, x_i) K(\alpha_k - \alpha_i, x_i)], \\ Q_2(\alpha_1, \alpha_2; \beta_1, \beta_2) &= H_{11}H_{22} + H_{12}H_{21} + \sum_p c_{4,p} + S_{\text{dbl}} \\ &\quad + \sum_{i,j} F(x_{ij}) [f_{i'j'} + K(\alpha_{i'} - \alpha_i, x_{ij}) K(\beta_{j'} - \beta_j, x_{ij})], \end{aligned}$$

where $x_{ij} = \alpha_i + \beta_j$, H is the identity part of G , D_4 and $c_{4,p}$ are the connected four-point terms of the independent-phase model, $f_{jk}^{(i)} = \sum_p \log^2 p \omega_p(x_i) \frac{q_j q_k}{(1-q_j)(1-q_k)}$ with $q_j = p^{-1-\alpha_j+\alpha_i}$ and $\omega_p(x) = \frac{(1-p^{-x})(1-p^{-1})p^{-x}(1-p^{-1-x})}{(1-2p^{-1}+p^{-1-x})^2}$, $f_{i'j'} = \sum_p \partial_{\alpha_{i'}} \partial_{\beta_{j'}} \log E_p$ is a closed form in the local factors (in the CUE it reduces to $H(x_{i'j'})$), and the double swap is $S_{\text{dbl}} = e^{-L(\alpha_1+\alpha_2+\beta_1+\beta_2)} \prod_{k,l} \zeta(1 \pm x_{lk}) / [\zeta(1 \pm (\beta_1 - \beta_2)) \zeta(1 \pm (\alpha_1 - \alpha_2))] A_{\text{dbl}}$. The Euler product for A_{dbl} converges only when all $x_{lk} < \frac{1}{2}$; outside that region we interpolate between its exact limits $A(x_{21})$ as $x_{12} \rightarrow 0$ and $A(x_{11})$ as $x_{22} \rightarrow 0$ (checked numerically), which moves κ_3 by at most $3 \cdot 10^{-4}$. Numerically we write $\kappa_3 = \kappa_3^{\text{CUE}}(L) + \iiint (I_\zeta - I_{\text{CUE}})$: the CUE part is the exact Palm value, and only the difference of the integrands is integrated numerically. This is a derivation under the ratios conjecture, with limits and integrals interchanged freely; it is not a theorem.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available at doi:10.5281/zenodo.22942475, the archive of the repository <https://github.com/azizran/riemann-zero-statistics> (directory `qm_riemann/`). It contains all code, data products, pre-registration files (with sha256 stamps and dates) and the research log, including the failed predictions; the LMFDB zero files are not redistributed, but their sha256 stamps are given. The ratios-conjecture predictions were committed before comparison to the author's private working repository (commits 182190f, bf214f0, a056eb0, e5f2580; post-audit correction 62b36dd). A file-by-file guide is in Section S8.

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